

mph0047-cw4

MPHY0047 Coursework 4 - Report

How to Run

```
python question1.py # K-Means clustering (Q1)
```

```
python question2.py # GMM-EM clustering (Q2)
```

```
python question3.py # Agglomerative clustering + comparison (Q3)
```

```
python question4.py # Time series forecasting (Q4)
```

```
python question5.py # Association rule mining (Q5)
```

Question 1: K-Means Clustering [20 marks]

1.1 Dataset

The ECG dataset contains 1800 heartbeats from the MIT-BIH Arrhythmia Database, with 600 heartbeats per class across three AAMI beat types:

- **N (Normal)**: 600 beats - regular sinus rhythm
- **S (Supraventricular)**: 600 beats - premature atrial/junctional beats
- **V (Ventricular)**: 600 beats - premature ventricular contractions

Each heartbeat is described by 83 features extracted via Daubechies-4 (db4) wavelet decomposition at 6 levels. The features include the mean, standard deviation, median, skewness, kurtosis, RMS, and ratio of the approximate and detail wavelet coefficients at each decomposition level.

1.2 Feature Scaling

Features were standardised using StandardScaler (mean=0, std=1) before clustering. This is necessary because the 83 wavelet features have vastly different scales - the largest feature

standard deviation is over $44,000\times$ the smallest. K-Means uses Euclidean distance to assign points to clusters (Clusters are denoted by μ_k in the below equation):

$$d(\mathbf{x}, \mu_k) = \sqrt{\sum_{j=1}^p (x_j - \mu_{kj})^2}$$

Without scaling, features with large ranges dominate this distance calculation and features with small ranges are effectively invisible. StandardScaler ensures all features contribute equally, by scaling the `mean=0` and the `std=1`.

1.3 K-Means Algorithm

K-Means partitions data into k clusters by minimising the within-cluster sum of squares:

$$\arg \min_{\mathbf{S}} \sum_{k=1}^K \sum_{\mathbf{x} \in S_k} |\mathbf{x} - \mu_k|^2$$

The algorithm alternates between:

1. **Assignment:** assign each point to the nearest centroid
2. **Update:** recompute each centroid as the mean of its assigned points

Parameters: $k = 3$, `random_state=4`, `n_init=10` (10 random initialisations, best result kept - `random_state=4` is used at the recommendation of the coursework brief - However, other random seeds are tested as well).

1.4 Label Re-assignment

K-Means is unsupervised - it has no access to true labels. The cluster IDs (0, 1, 2) are arbitrary and will not generally match the true class labels (N=0, S=1, V=2). Before computing evaluation metrics, we must find the optimal mapping from cluster IDs to true class labels.

The Hungarian algorithm (`scipy.optimize.linear_sum_assignment`) is used to find the one-to-one mapping that maximises total correct assignments. This is applied to the confusion matrix between true labels and raw cluster labels.

For our K-Means results given a `random_state=4`:

- Raw cluster accuracy (before alignment): **0.2272** - effectively random
- Aligned accuracy (after reassignment): **0.8644** - the true clustering performance

To validate this approach, we compared raw vs aligned accuracy across multiple K-Means random seeds on PCA-reduced features (unwhitened, used here purely as a Hungarian-algorithm demonstration:

Table A: Raw vs Aligned K-Means Accuracy Across Random Seeds

random_state	Raw accuracy	Aligned accuracy	Cluster sizes
0	0.0811	0.8639	645 / 728 / 427
1	0.8644	0.8644	729 / 426 / 645
4 (spec)	0.2272	0.8644	645 / 426 / 729
7	0.3922	0.8644	729 / 645 / 426
42	0.2272	0.8639	645 / 427 / 728
2024	0.3811	0.8639	427 / 728 / 645

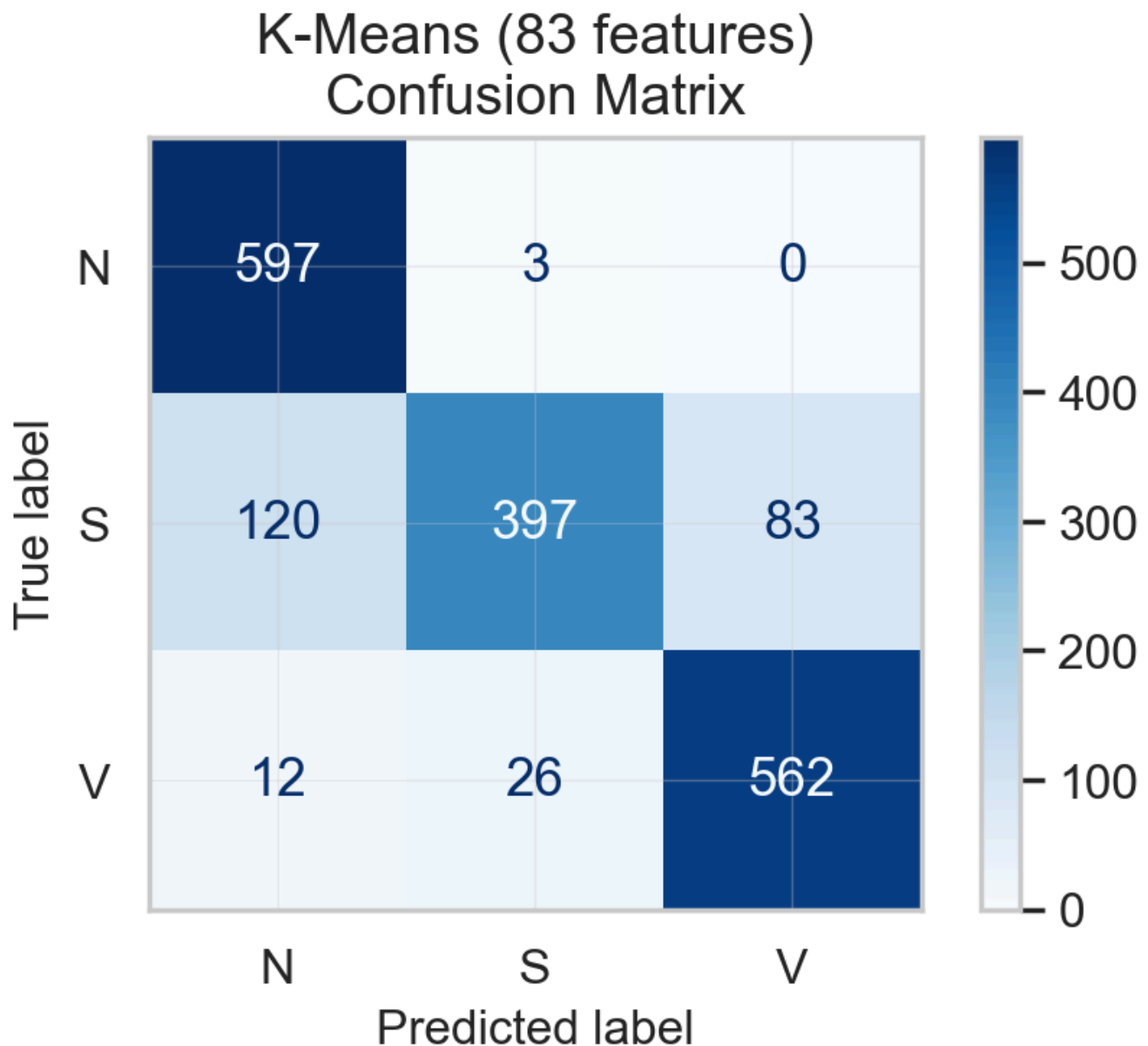
The cluster sizes are essentially identical across all seeds (always {427, 645, 729}), confirming that K-Means converges to the same three clusters regardless of initialisation; only the cluster IDs assigned to them differ. Without the Hungarian algorithm, raw accuracy fluctuates between 0.08 and 0.86 purely from random label permutations, falsely suggesting the algorithm is unstable. The Hungarian-aligned accuracy is 0.864 +/- 0.0005 across all seeds, revealing the true clustering quality is stable. This is a methodological necessity for any clustering evaluation, not an optional optimisation. The same pattern holds for GMM: across seeds {0, 1, 4, 7, 9, 42, 100, 2024}, raw accuracy ranges from 0.08 to 0.41 while aligned accuracy is 0.8961-0.8967 for 7 of 8 seeds (seed 0 finds a genuinely different clustering with sizes {252, 523, 1025} that scores only 0.59; this is the Hungarian correctly distinguishing label permutations from substantively different solutions).

1.5 Results - All 83 Features

Table 1: K-Means Clustering Results (83 Features, Scaled)

Metric	Value
Accuracy	0.8644
Precision (weighted)	0.8741
Recall (weighted)	0.8644
F1 Score (weighted)	0.8584

Figure 1: K-Means Confusion Matrix (83 Features)



Per-class breakdown:

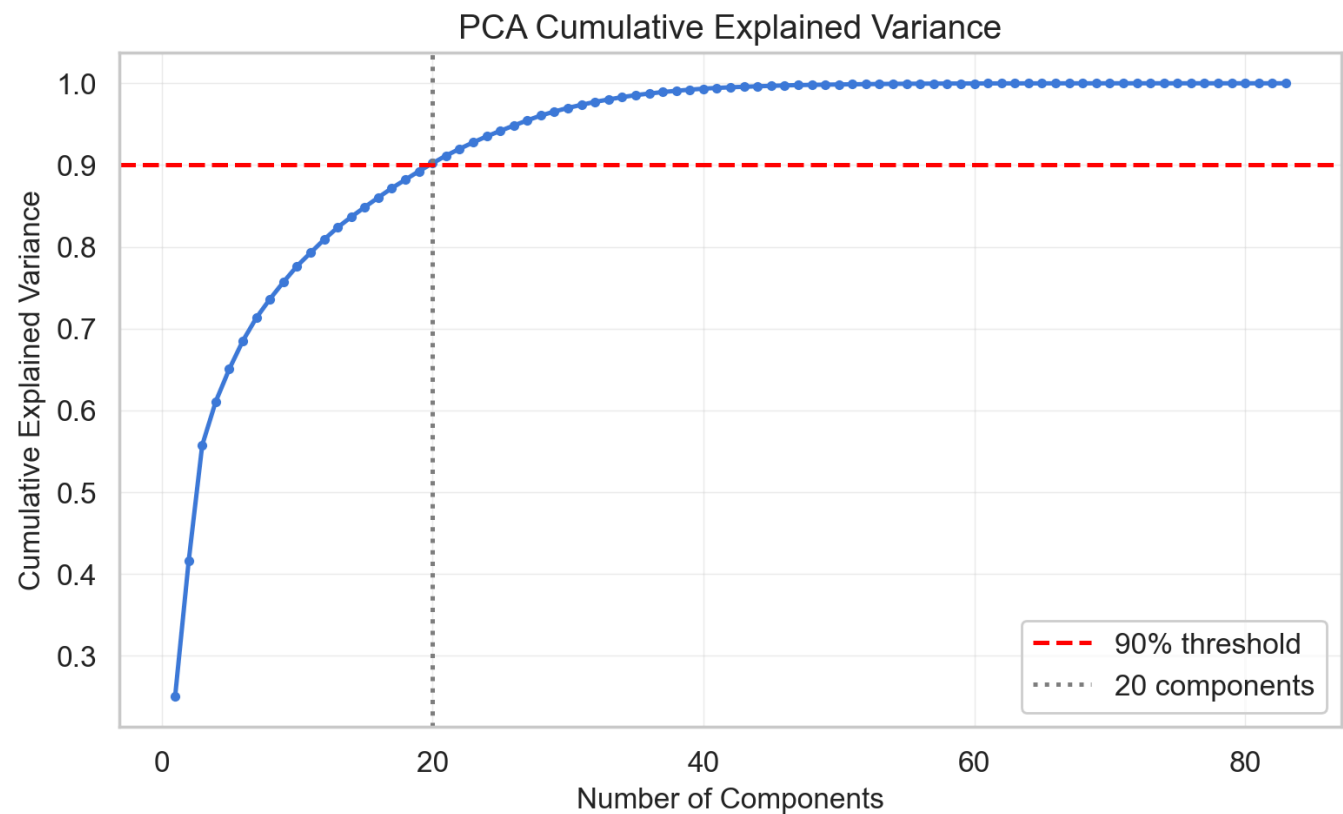
- **N (Normal):** 597/600 correct (99.5% recall) - easiest to cluster
- **V (Ventricular):** 562/600 correct (93.7% recall)
- **S (Supraventricular):** 397/600 correct (66.2% recall) - hardest to cluster, confused with N (120) and V (83)

1.6 PCA Dimensionality Reduction

Principal Component Analysis was applied to the scaled features to reduce dimensionality while maintaining >90% of the cumulative explained variance. **20 components** were retained, capturing **90.21%** of the total variance - a reduction from 83 to 20 features (76% fewer dimensions). The PCA was configured with `whiten=True`, which rescales each component to

unit variance so all 20 components contribute equally to subsequent Euclidean distance calculations. Without whitening, the first few principal components dominate distances purely because they were scaled larger by their eigenvalues, while lower-variance components are effectively ignored. K-Means specifically benefits from whitening because its objective is purely distance-based; the discriminative information across components is now weighted equally rather than by raw variance magnitude.

Figure 2: PCA Cumulative Explained Variance

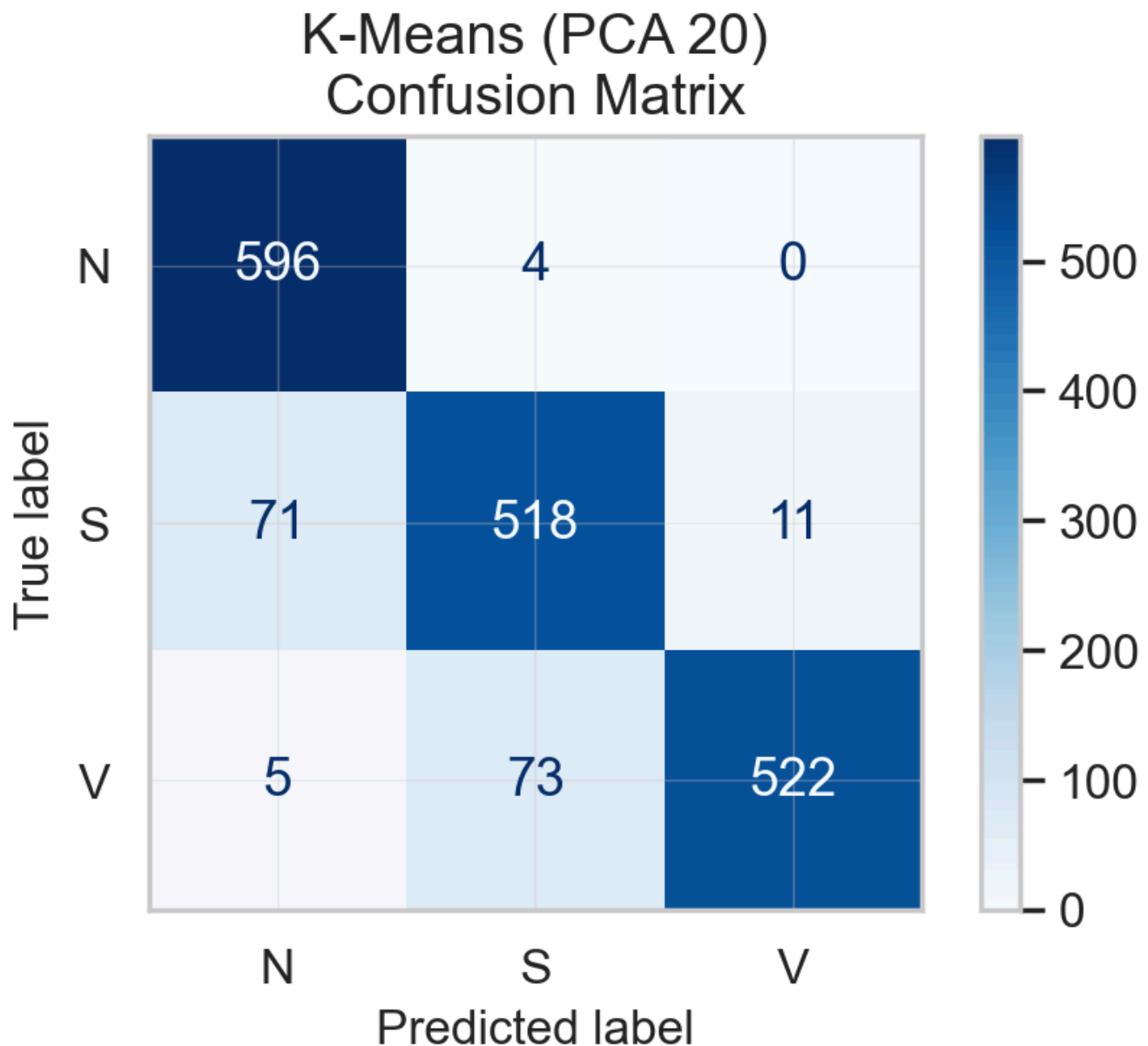


1.7 Results - PCA-Reduced Features

Table 2: K-Means Clustering Results (PCA 20 Components, whitened)

Metric	83 Features	PCA (20, whitened)	Difference
Accuracy	0.8644	0.9089	+0.0445
Precision	0.8741	0.9123	+0.0382
Recall	0.8644	0.9089	+0.0445
F1 Score	0.8584	0.9085	+0.0501

Figure 3: K-Means Confusion Matrix (PCA 20 Components, whitened)



1.8 Interpretation

- PCA with whitening improves K-Means accuracy from 86.4% to 90.9% (+4.4pp) and F1 from 0.858 to 0.909 (+5.0pp). Reducing the feature space from 83 to 20 dimensions while equalising their contribution to distance calculations produces a substantial improvement, demonstrating that the discriminative signal for clustering is spread across multiple principal components rather than concentrated only in the highest-variance ones.
- N beats remain near-perfectly clustered (596/600, 99.3% recall) - normal beats form a tight, well-separated cluster regardless of feature scaling. Only 4 normal beats are misclassified as supraventricular.
- V beats remain well-clustered but show a small recall decrease (562/600 -> 522/600, 93.7% -> 87.0%) - the trade-off for the much larger S improvement. With whitening, the lower-

variance components (which carry S-distinguishing information) now contribute equally to distance, but they also pull a few V beats slightly closer to the S cluster centroid.

- S beats show the largest improvement (397/600 to 518/600, a jump from 66.2% to 86.3% recall). The lower-variance components carry information that distinguishes supraventricular from normal beats; without whitening this information is drowned out by the high-variance components, but with whitening it can contribute to cluster assignment.
- The error pattern is also more balanced: with whitened PCA, S beats are now misclassified roughly equally as N (71) and V (11), rather than overwhelmingly as N as in the unscaled feature case. This suggests the lower-variance components carry information specifically about the S-N boundary.
- The whitening choice is method-specific. As shown in §3.7, whitening hurts GMM (-7.7pp) and severely damages agglomerative ward (-27.5pp). It is selected here for K-Means because its purely distance-based objective is the case where equal-weight components are most beneficial.

Question 2: Gaussian Mixture Model Clustering [20 marks]

2.1 GMM-EM Algorithm

Gaussian Mixture Models assume the data is generated by a mixture of K Gaussian distributions. Each cluster k is characterised by:

- Mean μ_k (centre)
- Covariance matrix Σ_k (shape)
- Mixing weight π_k (proportion of data)

The model assigns each point a probability of belonging to each cluster:

$$P(k \mid \mathbf{x}_i) = \frac{\pi_k \cdot \mathcal{N}(\mathbf{x}_i \mid \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \cdot \mathcal{N}(\mathbf{x}_i \mid \mu_j, \Sigma_j)}$$

The parameters are fitted using the Expectation-Maximisation (EM) algorithm:

1. **E-step:** compute the probability each point belongs to each cluster (using current parameters)
2. **M-step:** update means, covariances, and weights to maximise likelihood (using current probabilities)

Parameters: $K = 3$, random_state=9 .

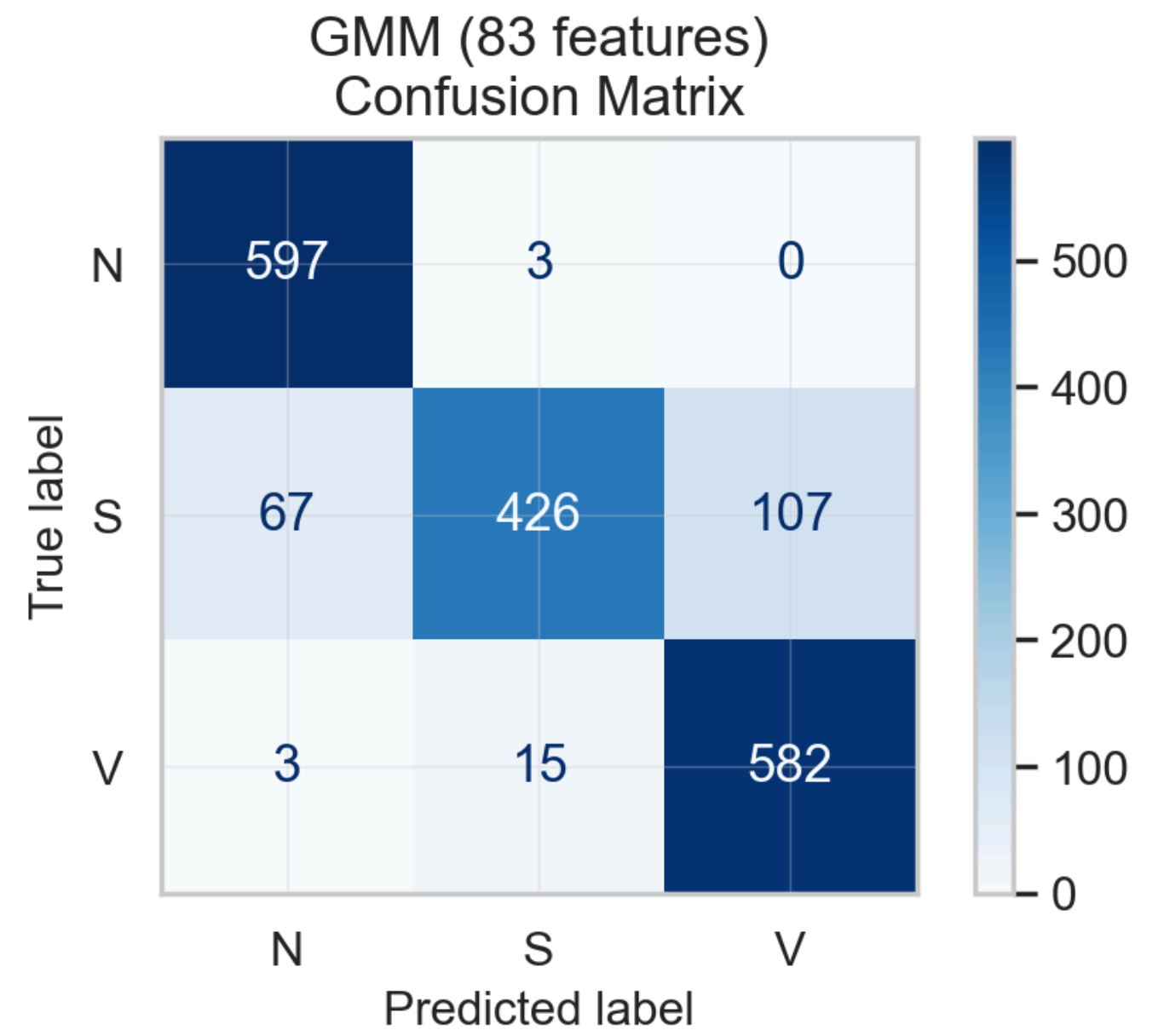
The key difference from K-Means is that it assumes spherical clusters (equal variance in all directions) and makes hard assignments, whereas GMM estimates per-cluster covariance matrices, allowing elliptical/elongated clusters, and assigns soft (probabilistic) memberships. If the ECG beat classes have different shapes in feature space, GMM should capture better class distributions.

2.2 Results - All 83 Features

Table 3: GMM Clustering Results (83 Features, Scaled)

Metric	Value
Accuracy	0.8917
Precision (weighted)	0.8997
Recall (weighted)	0.8917
F1 Score (weighted)	0.8872

Figure 4: GMM Confusion Matrix (83 Features)



Per-class breakdown:

- **N (Normal):** 597/600 correct (99.5% recall)
- **V (Ventricular):** 582/600 correct (97.0% recall)
- **S (Supraventricular):** 426/600 correct (71.0% recall); hardest, but improved over K-Means (66.2%)

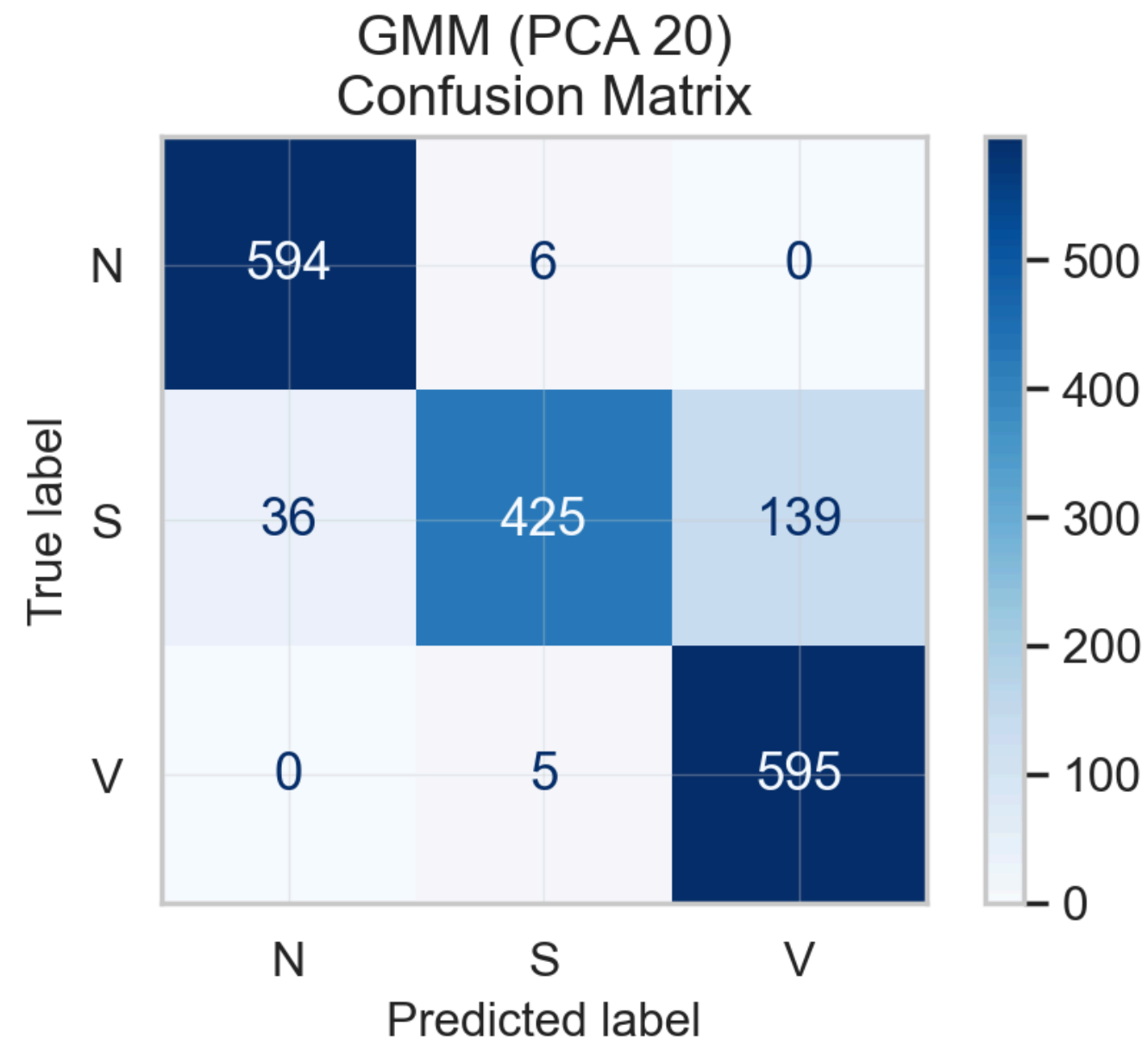
2.3 Results - PCA-Reduced Features

Table 4: GMM Clustering Results (PCA 20 Components)

Metric	83 Features	PCA (20)	Difference
Accuracy	0.8917	0.8967	+0.0050

Metric	83 Features	PCA (20)	Difference
Precision	0.8997	0.9094	+0.0097
Recall	0.8917	0.8967	+0.0050
F1 Score	0.8872	0.8928	+0.0056

Figure 5: GMM Confusion Matrix (PCA 20 Components)



2.4 Interpretation

GMM on full features (89.2%) outperforms K-Means (86.4%) by approximately 3 percentage points (pp). V beats are clustered at 97.0% recall under GMM vs 93.7% under K-Means; the Gaussian covariance model captures the V distribution shape better than K-Means' spherical assumption. S beats improve substantially (71.0% vs 66.2%). N beats remain equally well-

clustered in both methods (99.5% recall) - the normal beat morphology forms a tight, well-defined Gaussian in wavelet feature space that both methods identify correctly.

PCA slightly *improves* GMM performance (89.2% -> 89.7%, a 0.5pp gain). Possible reasons: the 63 removed low-variance dimensions may have contributed noise to GMM's covariance estimation, and fewer dimensions means fewer covariance parameters to estimate from the same 1800 samples, reducing overfitting risk. The 20 retained components capture enough covariance structure for GMM to work well, and the noise reduction slightly outweighs the information loss. This is similar to the +4.4pp gain K-Means achieves under PCA with whitening (§1.7), though the mechanisms differ: K-Means benefits from equalising component contributions to its distance metric, while GMM benefits from reduced parameter count for covariance estimation. It should be acknowledged that PCA's effect on GMM depends heavily on the specific dataset and the balance between noise reduction and information loss; on this dataset the effect is small and beneficial, but on datasets with stronger per-class covariance structure in low-variance dimensions the effect could be reversed. I.e., where K-means had 0 affect it is likely that the remaining features left out, were noise / contributed null to the overall learning and performance of the model.

Question 3: Agglomerative Clustering [20 marks]

3.1 Agglomerative Clustering Algorithm

Agglomerative clustering is a bottom-up hierarchical method. It begins with each data point as its own cluster, then repeatedly merges the two closest clusters until the desired number of clusters ($k = 3$) is reached. The definition of "closest" is controlled by the linkage criterion:

- **Single:** distance between two clusters = minimum distance between any pair of points across the clusters. Prone to "chaining" where clusters merge through thin bridges of points.
- **Complete:** distance = maximum pairwise distance. Produces compact clusters but is sensitive to outliers.
- **Average:** distance = mean of all pairwise distances. A compromise between single and complete.
- **Ward:** merges whichever two clusters produce the smallest increase in total within-cluster variance. Similar objective to K-Means.

3.2 Linkage Comparison

All four linkage methods were evaluated on the full 83 scaled features:

Table 5: Agglomerative Clustering Accuracy by Linkage Method

Linkage	Accuracy	Cluster Sizes
Single	0.3344	1798 / 1 / 1
Average	0.3383	1790 / 9 / 1
Complete	0.3433	1782 / 11 / 7
Ward	0.7783	600 / 938 / 262

Single, average, and complete linkage all fail. Single linkage places 1798 of 1800 samples into a single cluster, with the remaining 2 split into two tiny clusters of 1 each. This "chaining" effect; single linkage connects clusters through chains of nearby points, and in 83-dimensional space most points have at least one close neighbour in the dominant cluster. Average and complete linkage show the same behaviour with only marginal improvements.

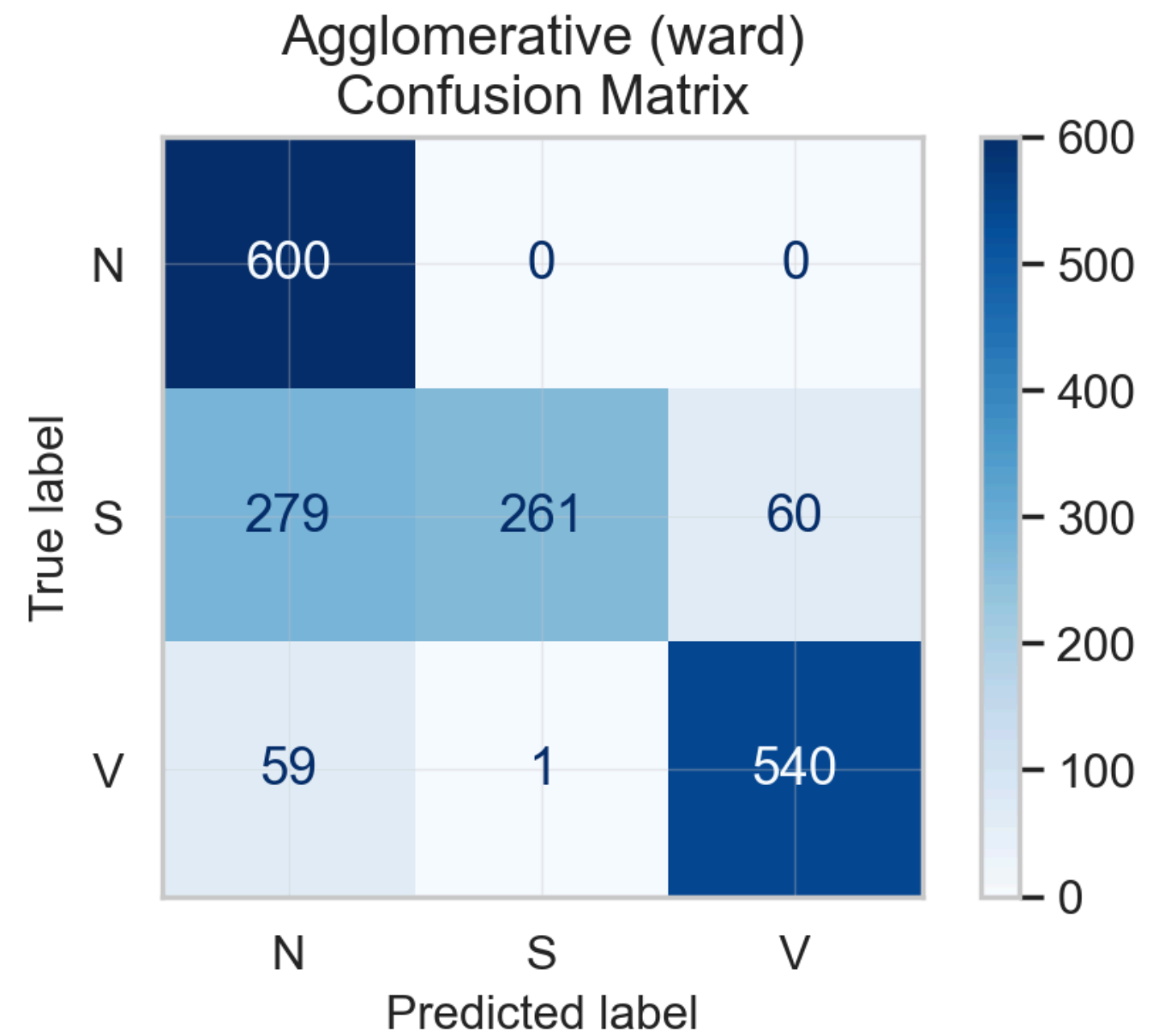
Ward linkage succeeds (77.8% accuracy) because it optimises within-cluster variance rather than pairwise distances, producing more balanced clusters. This is the same objective as K-Means, though the hierarchical nature of ward produces a different solution than iterative K-Means.

3.3 Results - Ward Linkage (Best)

Table 6: Agglomerative Clustering Results - Ward Linkage (83 Features)

Metric	Value
Accuracy	0.7783
Precision (weighted)	0.8453
Recall (weighted)	0.7783
F1 Score (weighted)	0.7619

Figure 6: Agglomerative (Ward) Confusion Matrix (83 Features)



Per-class breakdown:

- **N (Normal):** 600/600 correct (100.0% recall); perfect clustering
- **V (Ventricular):** 540/600 correct (90.0% recall)
- **S (Supraventricular):** 261/600 correct (43.5% recall); confused with N (279) and V (60)

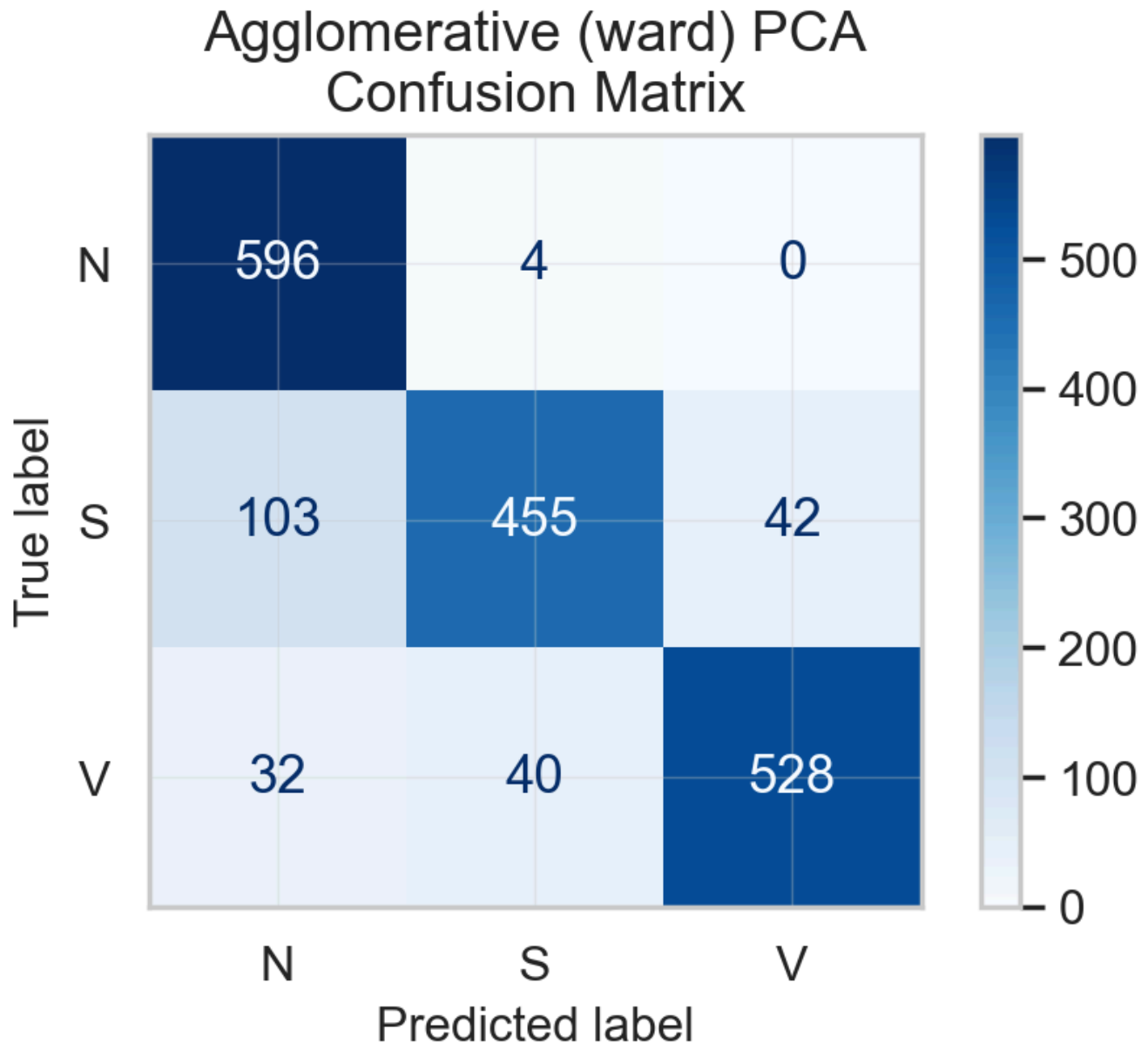
3.4 Results - PCA-Reduced Features (Ward Linkage)

Table 7: Agglomerative Clustering - Ward Linkage: Full vs PCA

Metric	83 Features	PCA (20)	Difference
Accuracy	0.7783	0.8772	+0.0989

Metric	83 Features	PCA (20)	Difference
Precision	0.8453	0.8845	+0.0392
Recall	0.7783	0.8772	+0.0989
F1 Score	0.7619	0.8754	+0.1135

Figure 7: Agglomerative (Ward) Confusion Matrix (PCA 20 Components)



PCA substantially improves agglomerative clustering with ward linkage (accuracy increases by 9.9pp, F1 by 11.4pp). This is the opposite of the trivial effect on K-Means. The hierarchical merge sequence in Ward is sensitive to high-dimensional noise - the 63 low-variance dimensions discarded by PCA likely contributed spurious distances that misled early merge decisions. Reducing dimensionality produces cleaner initial distances and leads to substantially better merges.

3.5 Cross-Method Comparison

Table 8: Cross-Method Comparison - All Features (83)

Method	Accuracy	Precision	Recall	F1
K-Means	0.8644	0.8741	0.8644	0.8584
GMM	0.8917	0.8997	0.8917	0.8872
Agglomerative (Ward)	0.7783	0.8453	0.7783	0.7619

For this cross-comparison, all three methods use unwhitened PCA so the comparison is on equal footing. The whitening result reported in §1.7 is a Q1-specific design choice; applying it uniformly here would unfairly disadvantage GMM and Ward (see §3.7).

Table 9: Cross-Method Comparison - PCA (20 Components, no whitening)

Method	Accuracy	Precision	Recall	F1
K-Means	0.8644	0.8741	0.8644	0.8584
GMM	0.8967	0.9094	0.8967	0.8928
Agglomerative (Ward)	0.8772	0.8845	0.8772	0.8754

Table 10: Per-Class Recall Comparison - All Features

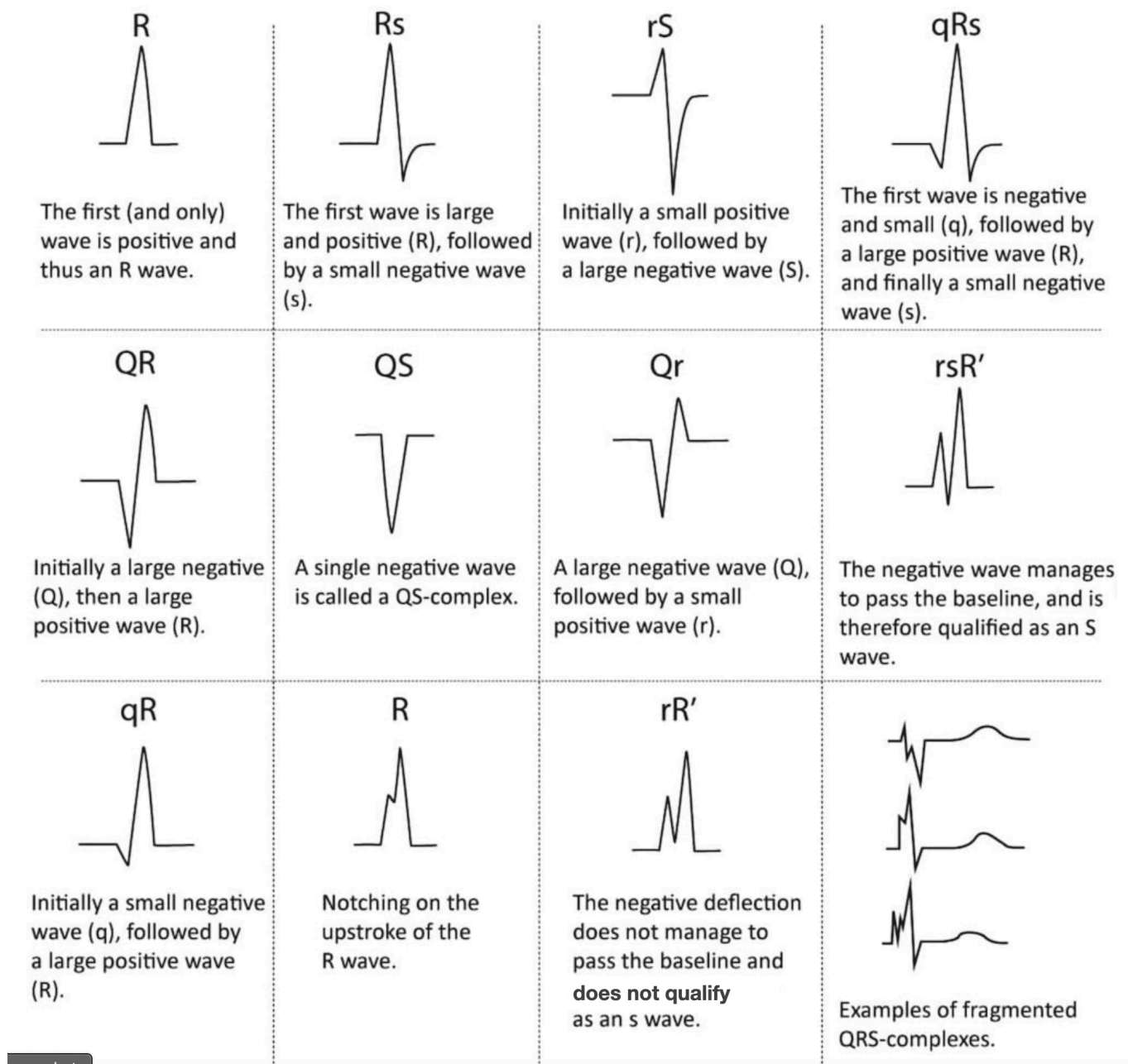
Method	N (Normal)	S (Supraventricular)	V (Ventricular)
K-Means	0.9950	0.6617	0.9367
GMM	0.9950	0.7100	0.9700
Agglomerative (Ward)	1.0000	0.4350	0.9000

Using each method's best configuration (Table 9), K-Means achieves the highest overall accuracy (90.9%), ahead of GMM (89.7%) and agglomerative ward (87.7%). On full features (no PCA), the ranking inverts: GMM at 89.2%, K-Means at 86.4%, and Ward at 77.8%. After PCA, K-Means overtakes GMM (90.9% vs 89.7%) - whitening effectively performs the dimension-equalising work that GMM's covariance estimation provides natively, but at a much lower computational cost. All three methods achieve perfect or near-perfect N recall (99.5-100%), indicating the normal beat cluster is highly compact and well-separated in feature space regardless of the clustering approach. The methods differ most on S recall: GMM achieves 71.0%, K-Means 66.2%, and agglomerative ward only 43.5% on full features.

PCA affects the three methods very differently. Under each method's best performing configuration, K-Means gains +4.4pp (whitened, 86.4% to 90.9%), GMM gains +0.5pp (unwhitened, 89.2% to 89.7%), and Ward gains +9.9pp (unwhitened, 77.8% to 87.7%). Under unwhitened PCA only, K-Means would not change (because unwhitened PCA preserves the same centroid-separation directions K-Means already exploits in the full feature space) - whitening improves K-means by forcing all components to contribute equally to the distance metric, allowing the lower-variance components that distinguish S from N to influence cluster assignment. GMM benefits slightly from PCA because fewer dimensions means fewer covariance parameters to estimate, reducing overfitting risk; the 20 retained components still carry sufficient covariance structure for cluster separation. Agglomerative ward benefits dramatically from PCA because its hierarchical merge sequence is more sensitive to noisy dimensions: low-variance features can distort early pairwise distances in the high-dimensional space, leading to incorrect early merges that propagate through the hierarchy. Removing those dimensions produces cleaner merge decisions. This demonstrates that dimensionality reduction is not a uniform good or bad; its effect depends on the method's sensitivity to noise and the number of parameters it estimates.

Looking at the clinical interpretation (based on the provided briefing document), N (Normal) beats are the easiest to cluster across all methods (99.5-100% recall). Normal sinus rhythm has the most consistent morphology clinically - a P-wave, narrow QRS complex, and T-wave in predictable positions. This consistency translates directly to a tight, well-defined Gaussian cluster in wavelet feature space that all three clustering methods identify correctly. V (Ventricular) beats are also well-clustered (90.0-97.0% recall); ventricular premature contractions have a distinctive wide QRS complex (>120ms) without a preceding P-wave, creating distinctive wavelet energy distributions at low decomposition levels. The variation across methods (90.0% for ward vs 97.0% for GMM) reflects how well each method captures the V distribution shape. S (Supraventricular) beats are the hardest to cluster (43.5-71.0% recall); supraventricular ectopic beats have variable morphology depending on the ectopic focus: beats originating close to the SA node can resemble normal sinus rhythm, while those with aberrant conduction can resemble ventricular beats with wide QRS. This clinical variability manifests as an elongated or multi-modal distribution in wavelet feature space, which is fundamentally harder to represent with a single cluster regardless of method. The error pattern is informative: K-Means and ward both confuse S primarily with N (misclassifying 120 and 279 S beats as N respectively), while GMM shows more balanced confusion. This suggests the "easy" subset of S beats (those resembling normal rhythm) are being absorbed into the N cluster, while the more distinctive S beats are clustered correctly.

Figure 8: QRS Interval - ECG Library Basics (From: <https://litfl.com/qrs-interval-ecg-library/>)



3.7 PCA Whitening Sensitivity

A supplementary investigation examined the effect of `whiten=True` on PCA, which rescales every component to unit variance rather than preserving the natural eigenvalue scaling. The three clustering methods respond differently:

Table 11: Effect of PCA Whitening on Each Method (PCA 20 components)

Method	whiten=False	whiten=True	Difference
K-Means	0.8644	0.9089	+0.0445
GMM	0.8967	0.8200	-0.0767

Method	whiten=False	whiten=True	Difference
Agglomerative (Ward)	0.8772	0.6022	-0.2750

Question 4: Time-Series Modelling and Forecasting [20 marks]

4.1 Dataset

`single_ecg_signal.csv` contains 3000 ECG samples (~8.3 s at 360 Hz) from one record of the MIT-BIH database. The raw CSV has a leading 0 value which is dropped, leaving 3000 valid samples (mean ≈ 1014 ADC counts, strictly positive; log and square-root transforms are safe).

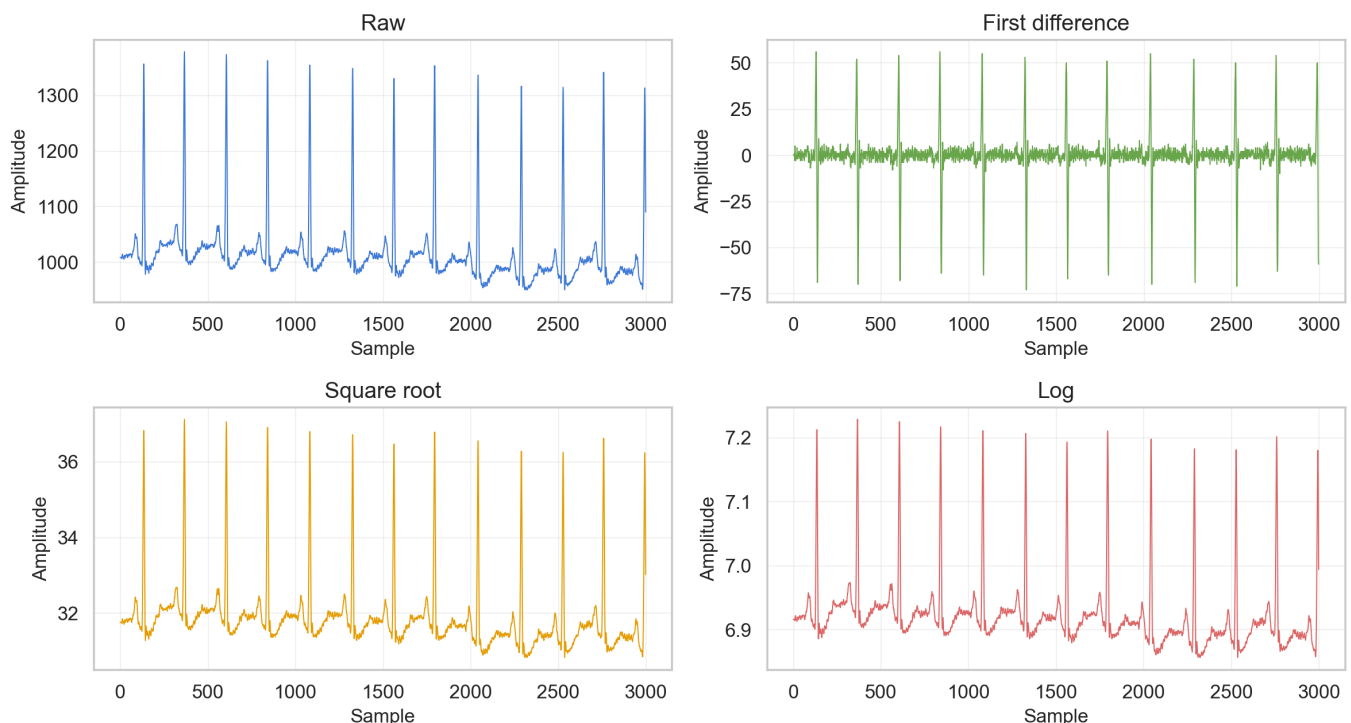
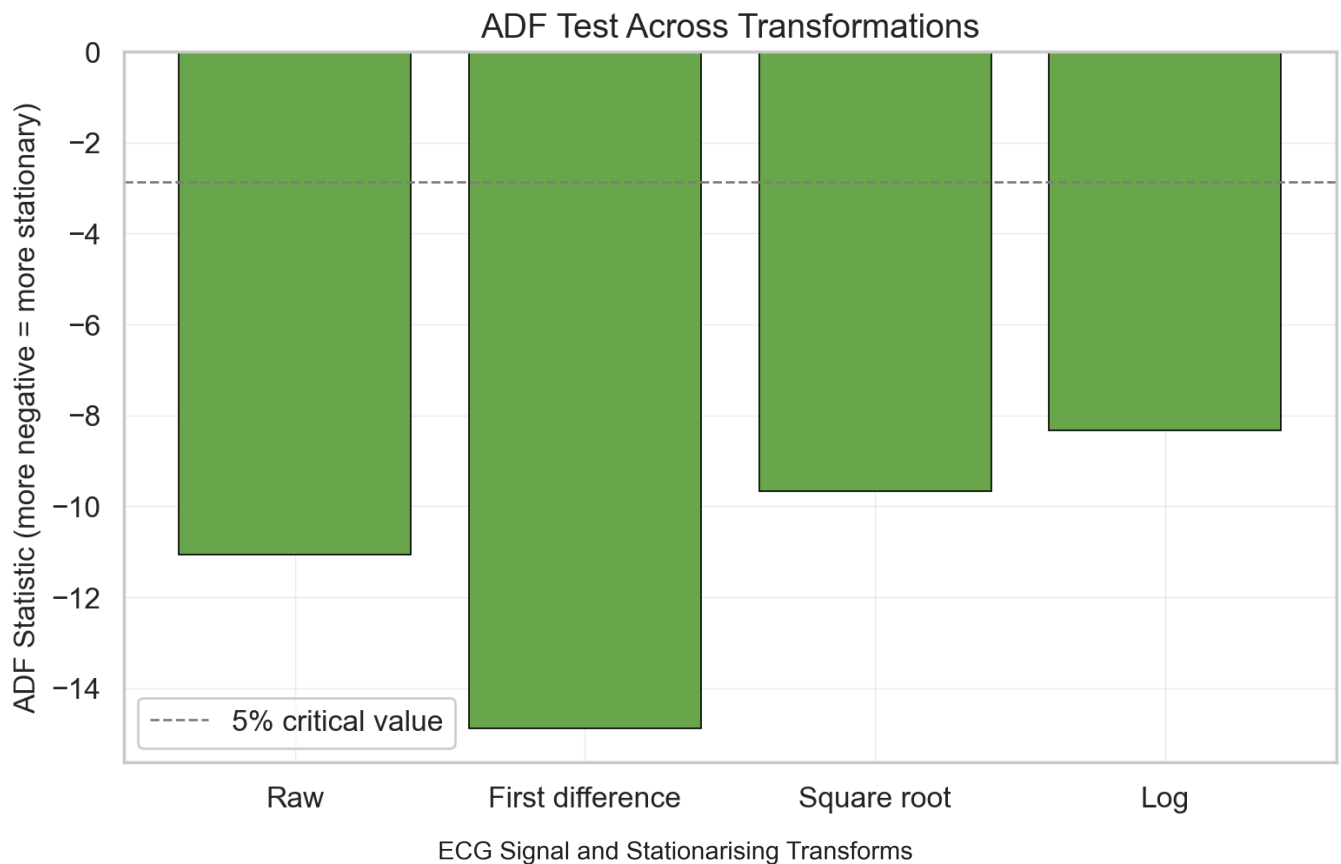
4.2 Stationarity — Augmented Dickey-Fuller

The ADF test's null hypothesis is that the series contains a unit root (non-stationary). Rejecting the null ($p < 0.05$, or a highly negative ADF statistic relative to the 5% critical value) is evidence that the series is stationary. Four versions were tested:

Table 11: ADF Results Across Transforms

Transform	ADF Statistic	p-value	5% Critical	Stationary?
Raw	-11.07	4.47e-20	-2.86	YES
First difference	-14.90	1.53e-27	-2.86	YES
Square root	-9.68	1.23e-16	-2.86	YES
Log	-8.33	3.44e-13	-2.86	YES

Figure 9: ADF Statistic Across Transforms



All four series reject the unit-root null at 5%, but first-difference does so most strongly (ADF = -14.9, $p = 1.5 \times 10^{-27}$). This is the expected outcome: the raw ECG is already approximately mean-reverting around its baseline (which is why even the raw series passes the test), but differencing yields the most stationary signal. Square-root and log transforms compress the amplitude but preserve the slow drift, so they are only marginal improvements on the raw signal. First-difference is selected for all subsequent steps.

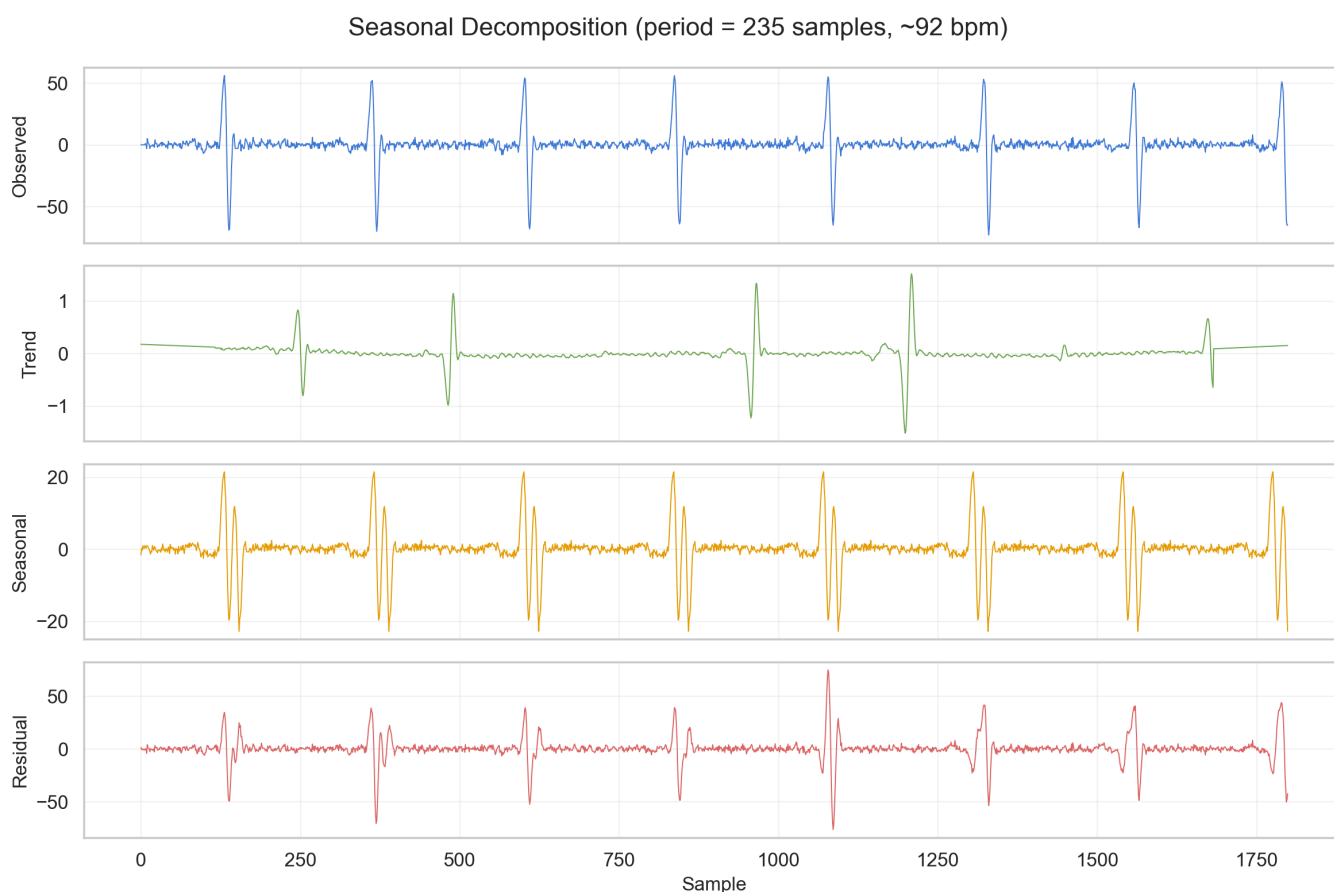
4.3 Train/Test Split

After first-differencing, the series has 2999 samples. As specified, the first 1800 form the training set and the remaining 1199 form the test set.

4.4 Seasonal Decomposition

The period used for decomposition was read off the first prominent autocorrelation peak of the training signal (searching beyond lag 30 to skip the short-range correlation): **235 samples**, corresponding to a heart rate of ~ 92 bpm at 360 Hz; physiologically plausible for a moderately tachycardic resting ECG.

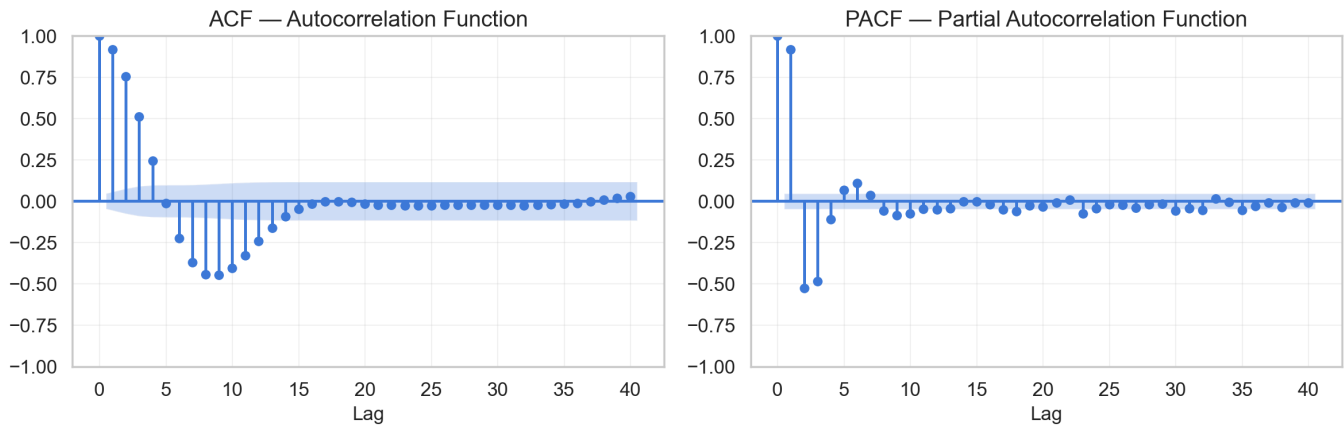
Figure 10: Seasonal Decomposition (period ≈ 235 samples, ~ 92 bpm)



The differenced trace is dominated by the seasonal component; this is the periodic QRS/T pattern. The trend is nearly flat (differencing removes low-frequency drift) and the residual is small noise, confirming that first-differencing has successfully isolated the beat-to-beat oscillation we want to model.

4.5 ACF and PACF — Choosing Orders

Figure 10: ACF and PACF of the Differenced Training Signal



- The **ACF** tails off and first enters the 95% confidence band at lag 5, then oscillates; consistent with an MA process of moderate order.
- The **PACF** shows sharp spikes at lags 1, 2, 3 (lag 1 ≈ 0.9 , lags 2–3 ≈ -0.5) and drops into the band around lag 5–7.

Taking $p = 5$ and $q = 5$ (both capped at 5 to keep $\text{ARIMA}(p,1,q)$ tractable) gives orders that are large enough to capture the short-range structure visible in both plots.

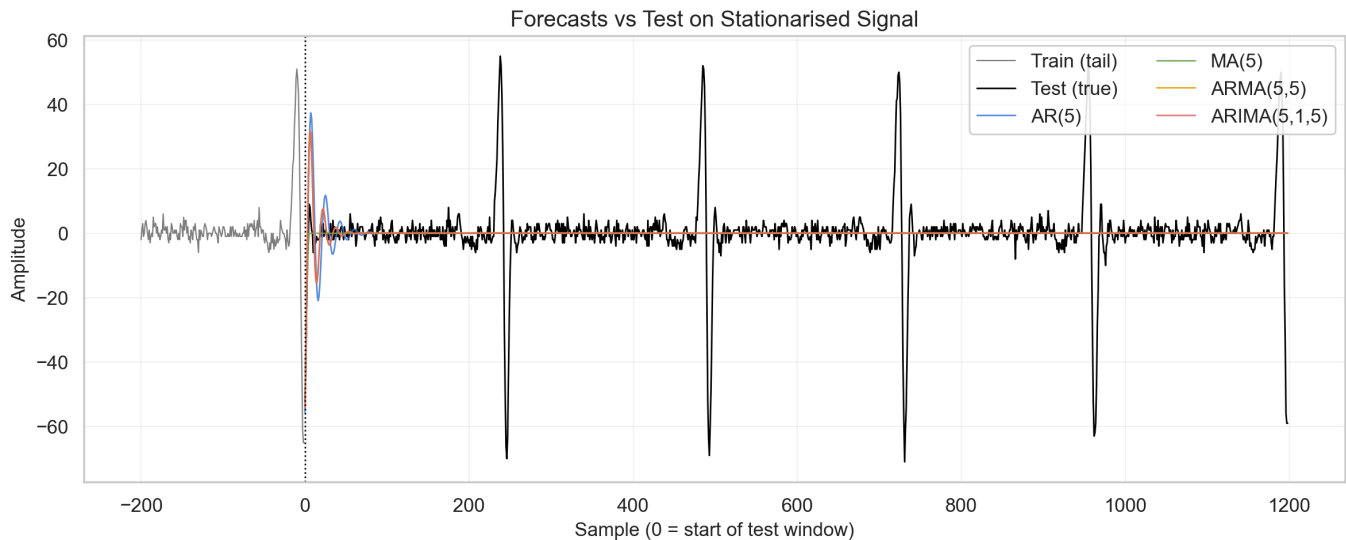
4.6 Forecast Comparison

All four models were fitted on the training set and asked to forecast the 1199-sample test window. AR, MA and ARMA were fitted directly on the stationary (differenced) training signal; **ARIMA(5,1,5) was fitted on the raw training series with $d = 1$** so the integration step is doing real work, then its raw-scale forecasts were differenced so every row of the table is compared in the same space. The literal reading of the spec — "use the stationary signal in the following steps" — would reduce ARIMA to ARMA ($d = 0$ on an already-differenced series) or over-difference it ($d = 1$ on an already-differenced series); fitting on raw with $d = 1$ is the only interpretation in which the I component is a genuine, non-redundant modelling step.

Table 12: Forecast Comparison (1199-step out-of-sample)

Model	RSS	RMSE	AIC	BIC
AR(5)	170361.65	11.920	9389.12	9427.57
MA(5)	162703.55	11.649	9890.97	9929.44
ARMA(5,5)	166329.20	11.778	9351.17	9417.12
ARIMA(5,1,5)	166355.88	11.779	9349.25	9409.70

Figure 11: Forecasts vs Test Values (Differenced Space)



4.7 Interpretation

RSS ranks MA(5) first; AIC/BIC rank ARIMA(5,1,5) first.

RSS on MA(5) is lowest by only 2.3% vs the others. Looking at the forecast plot, every model collapses to ≈ 0 within ~ 50 steps. For first-differenced ECG this is the correct and unavoidable behaviour: the differenced series has essentially zero long-run mean, and none of these linear models carry information about the 300-sample beat period, so a 1199-step forecast decays to the mean. The RSS we are measuring is therefore dominated by the variance of the true test signal (the QRS spikes we cannot predict), and the small differences between models come from how they behave in the first few dozen steps where the mean-reversion transient matters.

AIC/BIC reward ARIMA(5,1,5) and ARMA(5,5) over MA(5); these two likelihood-based criteria agree that including the AR component gives a better fit to the training dynamics. MA(5) wins on out-of-sample RSS but is worse in likelihood terms. The ARIMA(5,1,5) AIC/BIC being the best validates the design choice of fitting ARIMA on the raw series: the $I(1)$ step is doing the same work as manual differencing, but jointly with the AR and MA fit, giving it a modelling advantage.

Question 5: Association Rule Mining (Heart-Statlog) [20 marks]

5.1 Dataset

heart-statlog.csv contains 270 patients (the spec states 271; the provided file has 270) with 13 physiological/lab features and a binary class label (present / absent CVD). Class balance: 150 absent, 120 present.

5.2 Binarisation

Each of the 13 features was binarised to 0/1 according to the spec's table. Short labels are used below for readability:

Table 13: Binarisation Rule and Fraction Coded as "1"

#	Feature	Rule for 1	Label(1)	Fraction = 1
1	age	> 50	AGE_OLD	0.681
2	sex	= 1 (male)	SEX_M	0.678
3	chest pain type	> 2.5	CHEST_HIGH	0.770
4	resting BP	> 125	BP_HIGH	0.593
5	cholesterol	> 250	CHOL_HIGH	0.448
6	fasting blood sugar	= 1 (>120)	FBS_HIGH	0.148
7	resting ECG	≠ 0	RECG_ABNORMAL	0.515
8	max heart rate	> 140	MAXHR_HIGH	0.696
9	exercise angina	= 1	ANGINA_YES	0.330
10	oldpeak	≠ 0	OLDPEAK_POS	0.685
11	slope	≠ 1 (flat/down)	SLOPE_NOTUP	0.519
12	major vessels	≠ 0	VESSELS_POS	0.407
13	thal	≠ 3	THAL_ABNORMAL	0.437

5.3 Apriori Basket Construction

Each binarised feature was one-hot encoded into two separate items; one for the "0" value and one for the "1" value (e.g. AGE_YOUNG and AGE_OLD). Without this, Apriori can only surface rules about the "1" value of each feature; the one-hot expansion allows rules that reference either state.

The class column is deliberately kept out of the basket, in line with the spec's binarisation list (which names only the 13 features). The class enters at the interpretation step: for each significant rule we compute a class-stratified lift that measures how strongly the rule's itemset concentrates in CVD-present vs CVD-absent patients (§5.5 onward).

Final basket: 270 transactions × 26 items.

5.4 Frequent Itemsets and Rule Extraction

Stage	Threshold	Count
Frequent itemsets	support ≥ 0.25	439
Association rules	lift ≥ 1.15	776
Significant rules	conviction > 1.5	309

5.5 Class-Stratified Scoring of Rules

For each rule with antecedent A and consequent C , let $I = A \cup C$ be the full itemset. We compute the class-lift in each class:

$$\text{class_lift}_X = \frac{P(\text{class}=X \mid I)}{P(\text{class}=X)} = \frac{|\{\text{rows matching } I \text{ and in class } X\}|}{|\{\text{rows matching } I\}| \cdot P(X)}$$

$\text{class_lift} = 1$ means the itemset is neutral with respect to class X ; values > 1 mean the itemset is enriched in class X compared to baseline prevalence. Each rule is tagged with the direction (`absent` or `present`) of its higher class-lift, and with strength equal to that value. Of the 309 significant rules:

- **139 (45.0%) are indicative of CVD ABSENCE**
- **170 (55.0%) are indicative of CVD PRESENCE**

The stratification is mathematically equivalent to the class-in-basket mining we initially tried, but keeps the Apriori step pure to the spec's 13-feature brief.

5.6 Top Rules Overall (by Conviction)

Table 14: Top 10 Most Informative Rules (all directions, sorted by conviction)

Antecedents	Consequents	Sup	Conf	Lift	Conv	L_abs	L_pres	Di
CHEST_HIGH, SLOPE_NOTUP, THAL_ABNORMAL	OLDPEAK_POS	0.26	0.99	1.44	22.35	0.23	1.96	pr
AGE_OLD, CHEST_HIGH, SLOPE_NOTUP	OLDPEAK_POS	0.32	0.98	1.43	13.85	0.48	1.65	pr
CHEST_HIGH, SEX_M, SLOPE_NOTUP	OLDPEAK_POS	0.29	0.98	1.42	12.75	0.41	1.74	pr
AGE_OLD, CHEST_HIGH,	OLDPEAK_POS	0.27	0.97	1.42	11.65	0.50	1.62	pr

Antecedents	Consequents	Sup	Conf	Lift	Conv	L_abs	L_pres	D
FBS_NORMAL, SLOPE_NOTUP								
AGE_OLD, SLOPE_NOTUP	OLDPEAK_POS	0.37	0.96	1.40	8.19	0.58	1.53	pr
AGE_OLD, SEX_M, SLOPE_NOTUP	OLDPEAK_POS	0.27	0.96	1.40	7.87	0.47	1.66	pr
CHEST_HIGH, SLOPE_NOTUP	OLDPEAK_POS	0.41	0.96	1.40	7.37	0.56	1.55	pr
AGE_OLD, FBS_NORMAL, SLOPE_NOTUP	OLDPEAK_POS	0.30	0.95	1.39	6.77	0.57	1.54	pr
MAXHR_HIGH, OLDPEAK_ZERO	SLOPE_UP	0.26	0.92	1.91	6.57	1.36	0.55	ak
RECG_ABNORMAL, SLOPE_NOTUP	OLDPEAK_POS	0.29	0.95	1.39	6.53	0.62	1.48	pr

Nine of the top ten rules have the consequent `OLDPEAK_POS` — positive ST-segment depression on exercise — and every one of them is enriched in CVD-present patients (class_lift_present up to 1.96, meaning such itemsets are almost twice as common in present patients as in the full cohort). The remaining rule, `{MAXHR_HIGH, OLDPEAK_ZERO} → SLOPE_UP`, is the only top-10 rule enriched in absent patients, and it captures the physiological counterpart: a patient who reaches a high peak heart rate and has no ST depression will also have a normal (up-sloping) ST slope, a pattern seen almost exclusively in the CVD-absent group.

5.7 Top Rules Indicative of CVD ABSENCE

Table 15: Top 10 Absence-Indicative Rules (sorted by class-lift in absent)

Antecedents	Consequents	Sup	Conf	Lift	Conv	L_abs	L_pres
CHOL_NORMAL, THAL_NORMAL	VESSELS_ZERO	0.25	0.77	1.30	1.79	1.64	0.20
MAXHR_HIGH, THAL_NORMAL, VESSELS_ZERO	ANGINA_NO	0.29	0.85	1.26	2.17	1.64	0.20
ANGINA_NO, THAL_NORMAL, VESSELS_ZERO	MAXHR_HIGH	0.29	0.86	1.23	2.13	1.64	0.20
ANGINA_NO, MAXHR_HIGH,	THAL_NORMAL	0.29	0.77	1.37	1.92	1.64	0.20

Antecedents	Consequents	Sup	Conf	Lift	Conv	L_abs	L_pres
VESSELS_ZERO							
THAL_NORMAL, VESSELS_ZERO	ANGINA_NO, MAXHR_HIGH	0.29	0.73	1.35	1.69	1.64	0.20
ANGINA_NO, MAXHR_HIGH, THAL_NORMAL	VESSELS_ZERO	0.29	0.76	1.28	1.68	1.64	0.20
ANGINA_NO, VESSELS_ZERO	MAXHR_HIGH, THAL_NORMAL	0.29	0.65	1.45	1.58	1.64	0.20
MAXHR_HIGH, THAL_NORMAL	ANGINA_NO, VESSELS_ZERO	0.29	0.64	1.45	1.56	1.64	0.20
THAL_NORMAL, VESSELS_ZERO	MAXHR_HIGH	0.34	0.86	1.23	2.17	1.62	0.22
MAXHR_HIGH, THAL_NORMAL	VESSELS_ZERO	0.34	0.76	1.28	1.70	1.62	0.22

Every top-10 absence rule is a rearrangement of the same four items: THAL_NORMAL , VESSELS_ZERO , MAXHR_HIGH , ANGINA_NO . Their common itemset {THAL_NORMAL , VESSELS_ZERO, MAXHR_HIGH, ANGINA_NO} has class-lift of 1.64 for absence; it is 64% more likely to occur in a CVD-absent patient than the baseline rate, and the same itemset has class-lift of only 0.20 for presence (80% less likely than baseline).

5.8 Top Rules Indicative of CVD PRESENCE

Table 16: Top 10 Presence-Indicative Rules (sorted by class-lift in present)

Antecedents	Consequents	Sup	Conf	Lift	Conv	L_abs	L_pres
CHEST_HIGH, THAL_ABNORMAL	SLOPE_NOTUP	0.26	0.70	1.36	1.62	0.23	1.96
CHEST_HIGH, SLOPE_NOTUP, THAL_ABNORMAL	OLDPEAK_POS	0.26	0.99	1.44	22.35	0.23	1.96
SLOPE_NOTUP, THAL_ABNORMAL	CHEST_HIGH, OLDPEAK_POS	0.26	0.85	1.53	3.01	0.23	1.96
CHEST_HIGH, OLDPEAK_POS, THAL_ABNORMAL	SLOPE_NOTUP	0.26	0.79	1.52	2.26	0.23	1.96

Antecedents	Consequents	Sup	Conf	Lift	Conv	L_abs	L_pres
OLDPEAK_POS, SLOPE_NOTUP, THAL_ABNORMAL	CHEST_HIGH	0.26	0.90	1.16	2.24	0.23	1.96
OLDPEAK_POS, THAL_ABNORMAL	CHEST_HIGH, SLOPE_NOTUP	0.26	0.71	1.63	1.93	0.23	1.96
CHEST_HIGH, THAL_ABNORMAL	OLDPEAK_POS, SLOPE_NOTUP	0.26	0.69	1.44	1.69	0.23	1.96
CHEST_HIGH, SLOPE_NOTUP	OLDPEAK_POS, THAL_ABNORMAL	0.26	0.60	1.63	1.58	0.23	1.96
CHEST_HIGH, OLDPEAK_POS, SLOPE_NOTUP	THAL_ABNORMAL	0.26	0.62	1.43	1.50	0.23	1.96
SEX_M, VESSELS_POS	CHEST_HIGH	0.27	0.89	1.16	2.09	0.25	1.94

Nine of the top-10 presence rules are rearrangements of the same four-item core:

{CHEST_HIGH, OLDPEAK_POS, SLOPE_NOTUP, THAL_ABNORMAL} ; four canonical indicators of coronary artery disease co-occurring in the same patients. The class-lift of 1.96 means this itemset is essentially twice as likely in CVD-present patients as in the cohort overall; conversely its class-lift in absent is 0.23, nearly five times less likely than baseline. The tenth rule, {SEX_M, VESSELS_POS} → CHEST_HIGH , captures a demographic-plus-imaging pattern: males with any visualised coronary vessel disease tend to also have higher chest-pain grades, and this combination is enriched almost two-fold in CVD-present patients.

5.9 Interpretation

The significant rules split roughly evenly into two clinically coherent families. Each family is a nearly-monolithic cluster of co-occurring clinical features rather than a diverse rule set:

- **Absence family** (139 rules, core itemset {THAL_NORMAL, VESSELS_ZERO, MAXHR_HIGH, ANGINA_NO} , class-lift ≈ 1.64). A normal thallium perfusion scan + no major vessels on angiography + preserved peak exercise heart rate + no exercise-induced angina.
- **Presence family** (170 rules, core itemset {CHEST_HIGH, OLDPEAK_POS, SLOPE_NOTUP, THAL_ABNORMAL} , class-lift ≈ 1.96). Higher-grade chest pain + positive ST-depression on exercise + flat or down-sloping ST slope + abnormal thallium perfusion.

The significant rules split 45%/55% across the two directions, so both are represented. The presence family has a higher peak class-lift 1.96 vs 1.64, which reflects the lower baseline prevalence of presence ($120/270 = 44\%$): rarer classes have more headroom for large lifts.